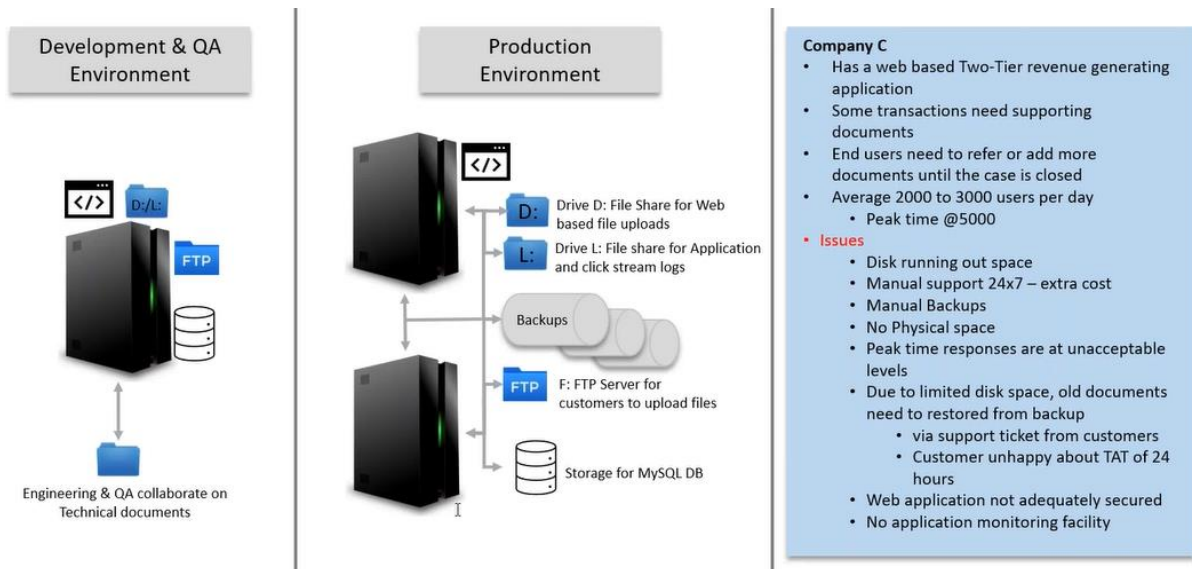


Migrating application to cloud

- Data
- Application
- Database

AWS Services: As interpreter from A to B, knowledge of AWS service is must to migrate.

Use Case:



AWS Data Migration Services:



Challenges

Moving terabytes or even petabytes of data to the cloud is a big challenge. This is a time-consuming task to move all that data to cloud. While companies plan and migrate this large scale of data to cloud, following factors must be considered seriously.

Data Storage

Pushing data into cloud without considering how that is going to be used. Do not consider cost alone as a primary factor, consider its usage too. Object storage(S3) vs File storage (EBS, EFS/FSx) needs careful consideration basis how they will be used by application. Consider changing File storage be Object storage after a period of time? This may be different for each use case.

Data Preparation

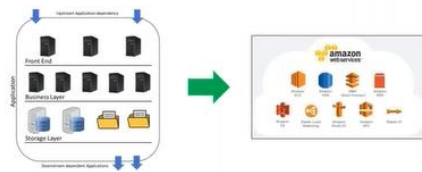
Filter out unwanted data. Prioritize what data should move first basis application that needs it. Same data may be required in different format when in cloud. Compression and archiving of data, aging definition or retention period for each dataset that is moving to cloud.

Validation

Integrity checking is the single most important step, and the easiest to get wrong. Use pessimistic approach, assume corruption will occur during data transfer. Employ checksums before and after. Do these checks before migrating applications/systems are turned off for good.

Transfer Marshalling

If data preparation involves copying, exporting, reformatting, or archiving, local storage can become a bottleneck. It may be necessary to set up dedicated local storage to stage the prepared data. This allows many systems to perform data preparation in parallel before the media is shipped.



The common cloud data migration challenge

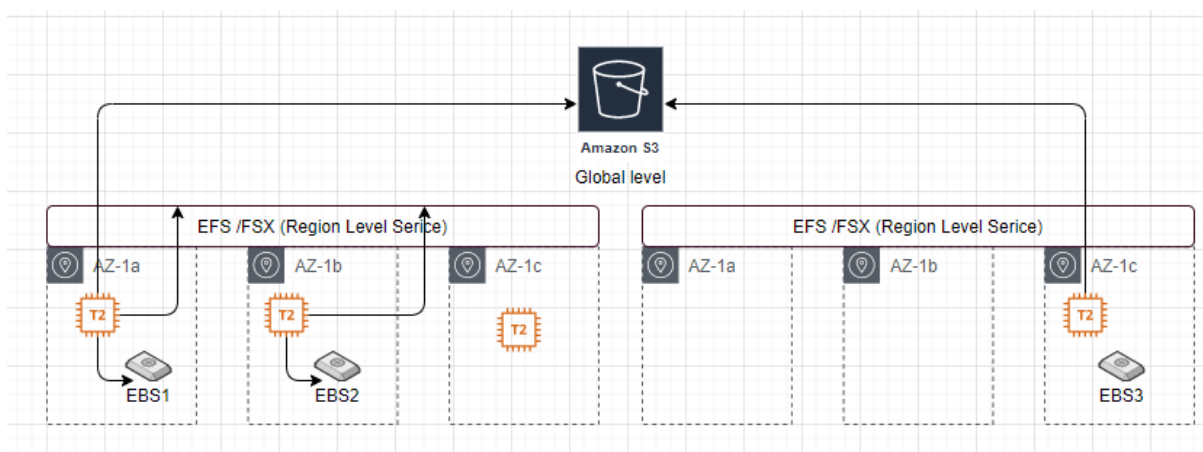
Type	Speed in mbps
T1	1.544
T2	6.312
T3	44.736

Data Storage:

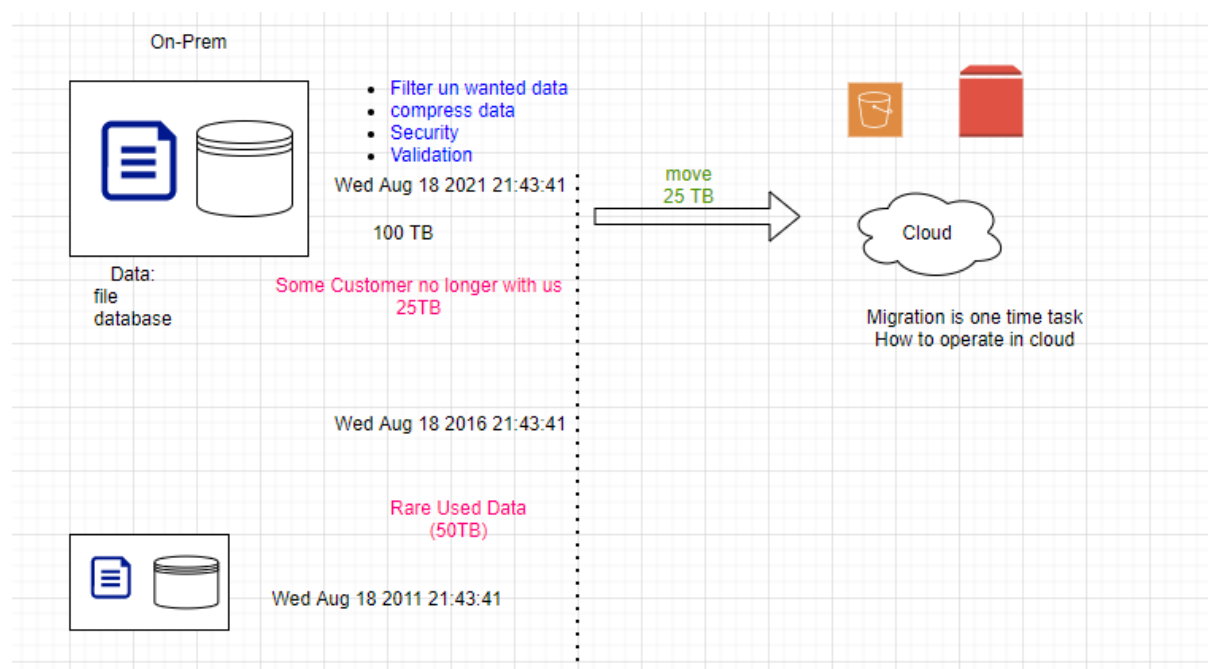
File Storage:

- Update
- Append
- Collaborate with other team members

Object Storage: can't do anything, its key value database when we update it overwrite, file name is key and its contents is value.



Data Preparation:



Estimate time to migrate, in range (30-40) not estimate (35).

Number of days = (Total Bytes)/(Megabits per second * 125 * 1000 * Network Utilization * 60 seconds * 60 minutes * 24 hours)

For example, if you have a T1 connection (1.544Mbps) and 1TB (1024 * 1024 * 1024 * 1024 bytes) to move in or out of AWS the theoretical minimum time it would take to load over your network connection at 80% network utilization is 82 days.

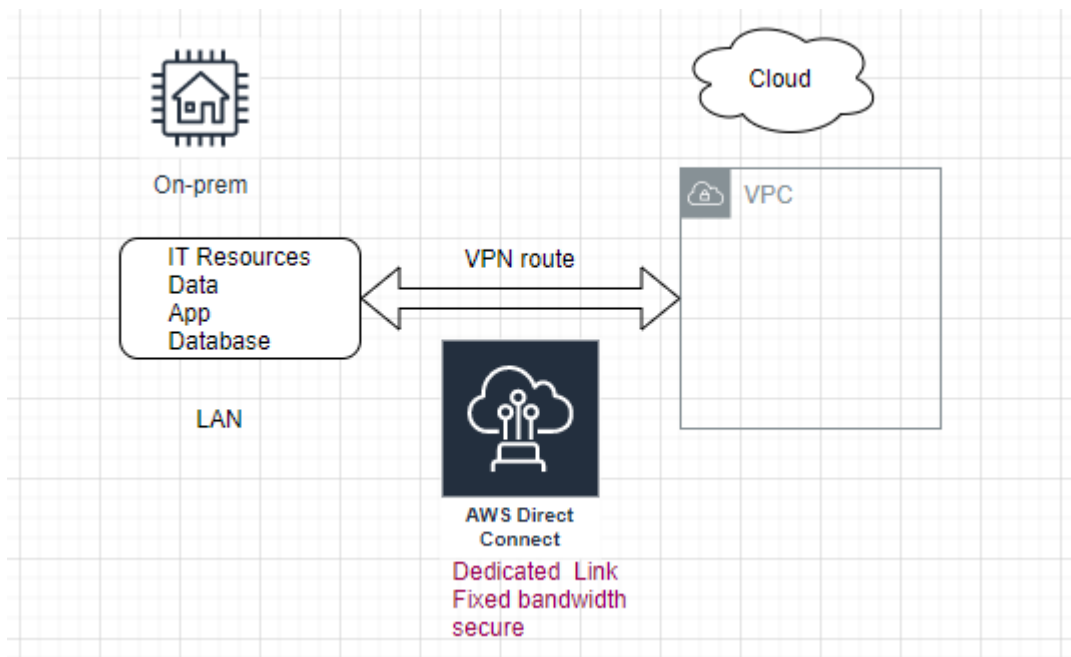
How to ensure the connectivity between the on-prem and cloud?

What is Gateway: Enable you to connect two different environments.

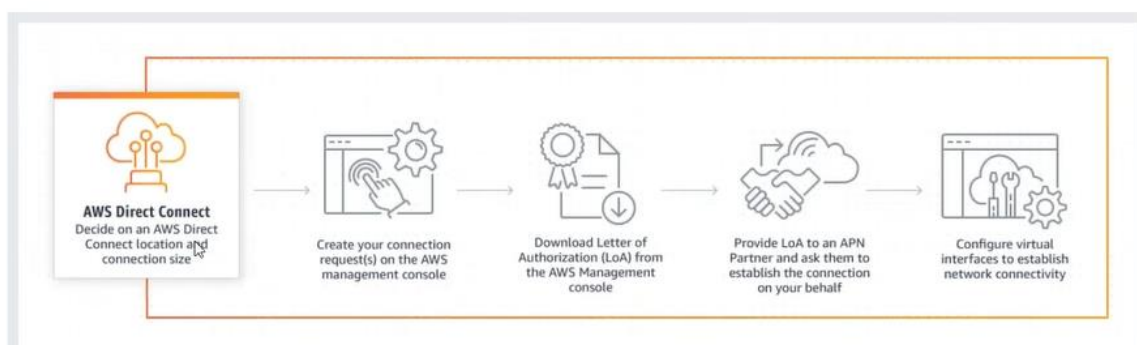
- Direct Connect
- Virtual private gateway
- Transit gateway

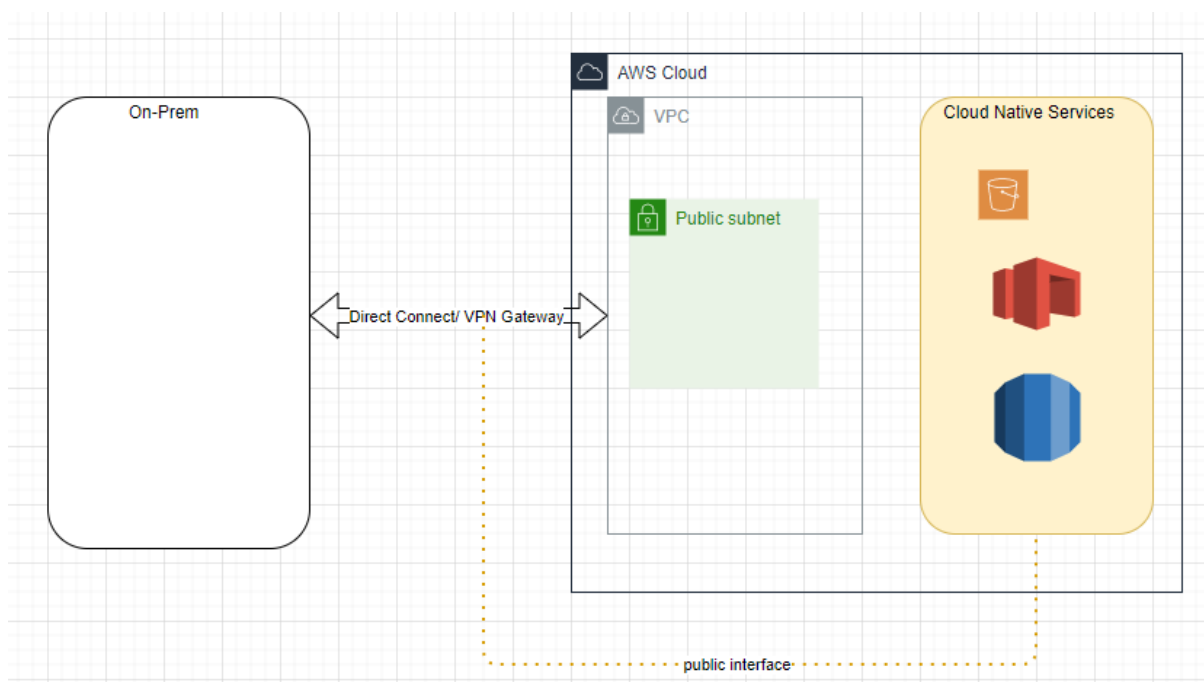
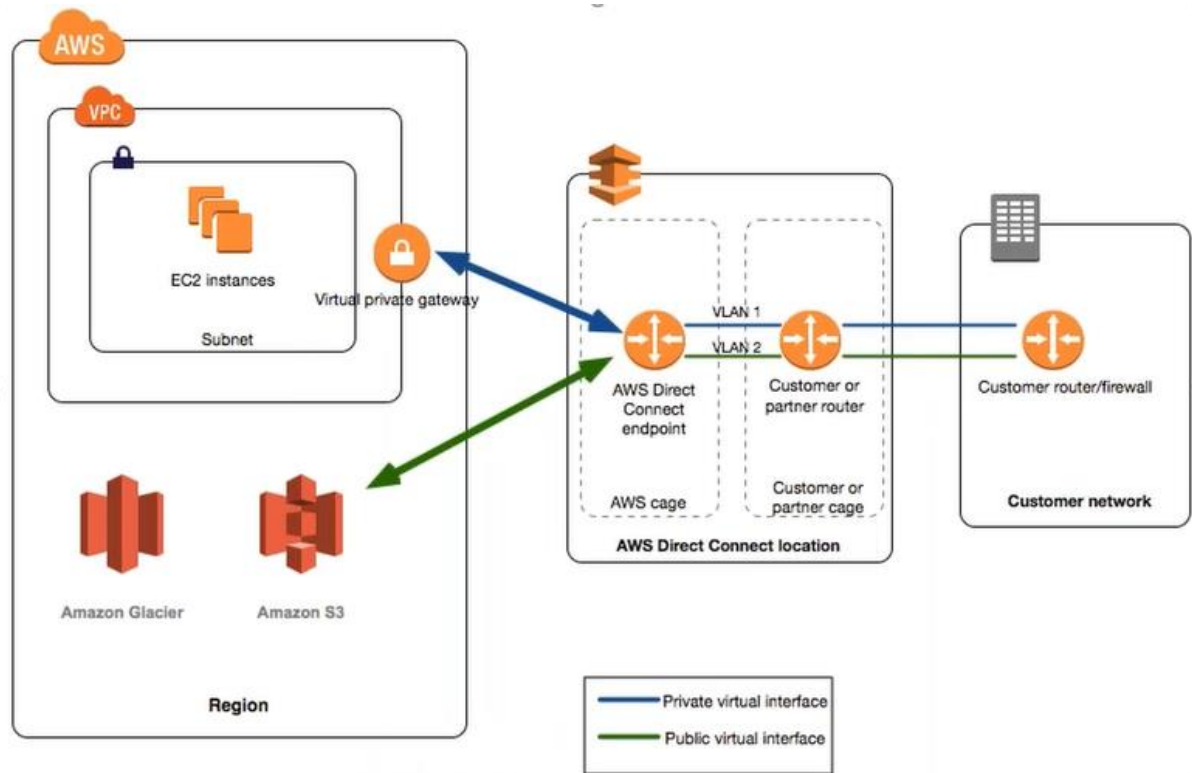
Direct Connect:

AWS Direct Connect makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect you can establish private network connectivity between AWS and your datacentre, office or colocation environment.



How it works





Direct connect can access VPC as well as Cloud Native Services.

Site-to-Site VPN Connectivity: If you want temporary connectivity and data is less than 500 GB, then go with VPN. If want to operate on Hybrid mode for too long then go for Direct Connect.

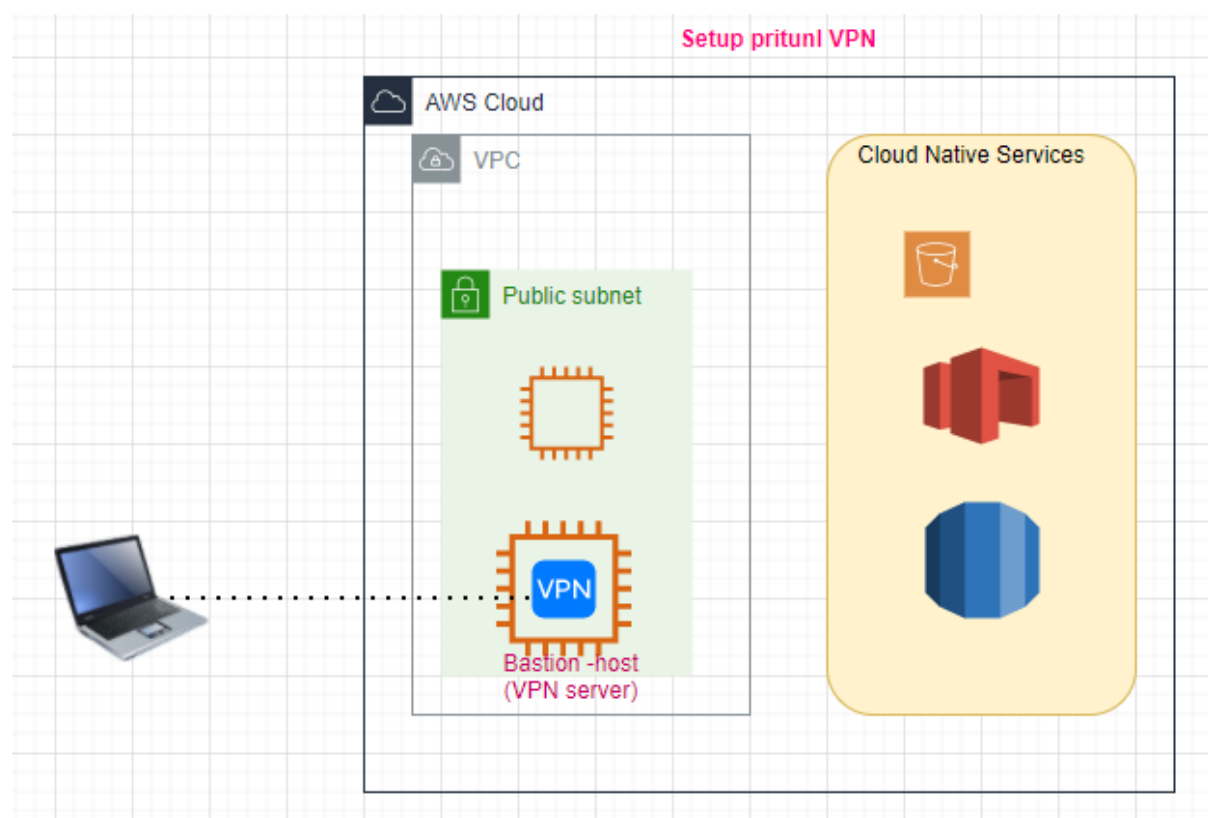
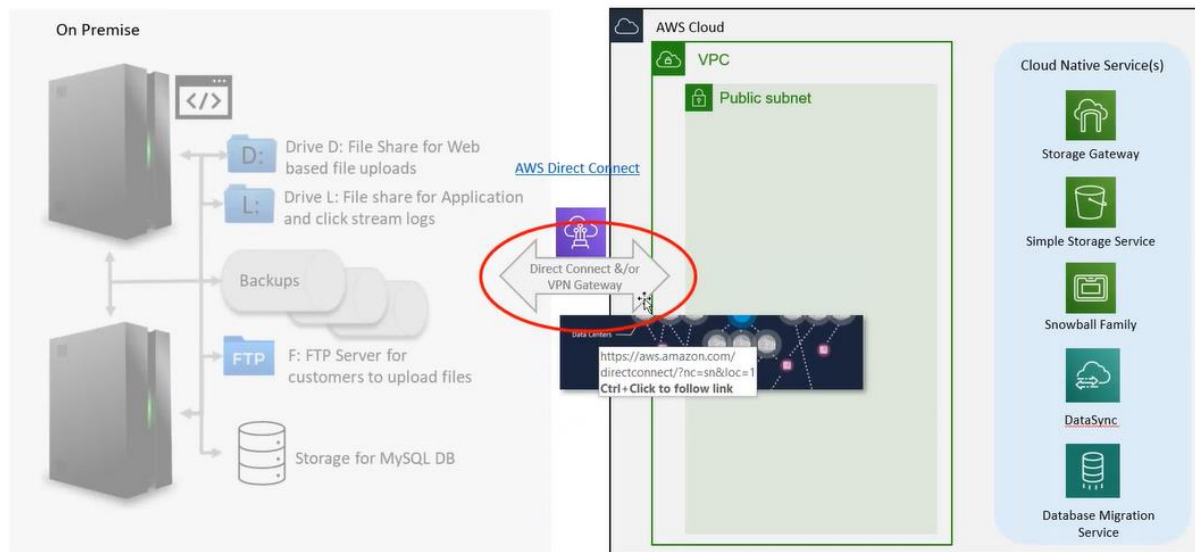
Virtual Private Gateway (Direct Connect): connects your VPC to the on-premises. Is secure but not encrypted.

Virtual Private Gateway is only to access your VPC level services, not to access cloud native services.

Enable VPN on top of Direct Connect. Direct Connect data flow through the third-party connectivity.

UC:

Once Connection establishes, either use machine on-prem or use machine on vpc both are same, they are part of same network.



step1: Launch an EC2 instance (bastion host)

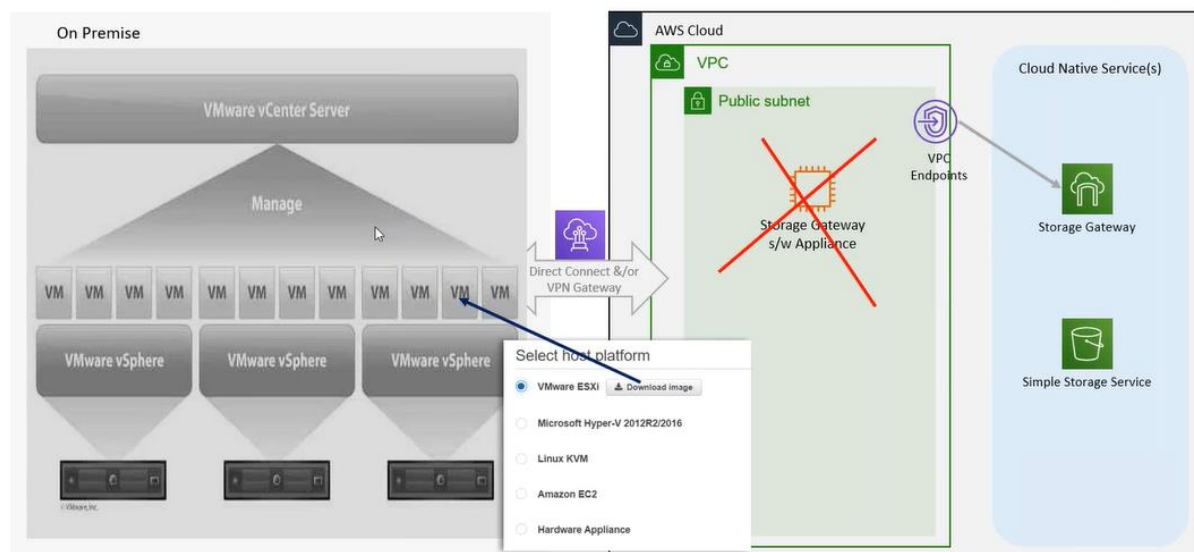
1:30

step2: Install pritunl s/w on EC2

Storage Gateway:

Cloud Coming to your on-prime, connect to the drives (D:/, E:/) in backend it stores in S3, translating of file-storage to object storage done through the s/w appliance.

step1: Create a Storage Gateway (Gateway type: Amazon S3 File Gateway).



step2: Platform options (EC2 m4. large + 50 GB).

step3: SG for storage gateway appliance.

Assign a security group: ☒ Create a new security group
☐ Select an existing security group

Security group name:

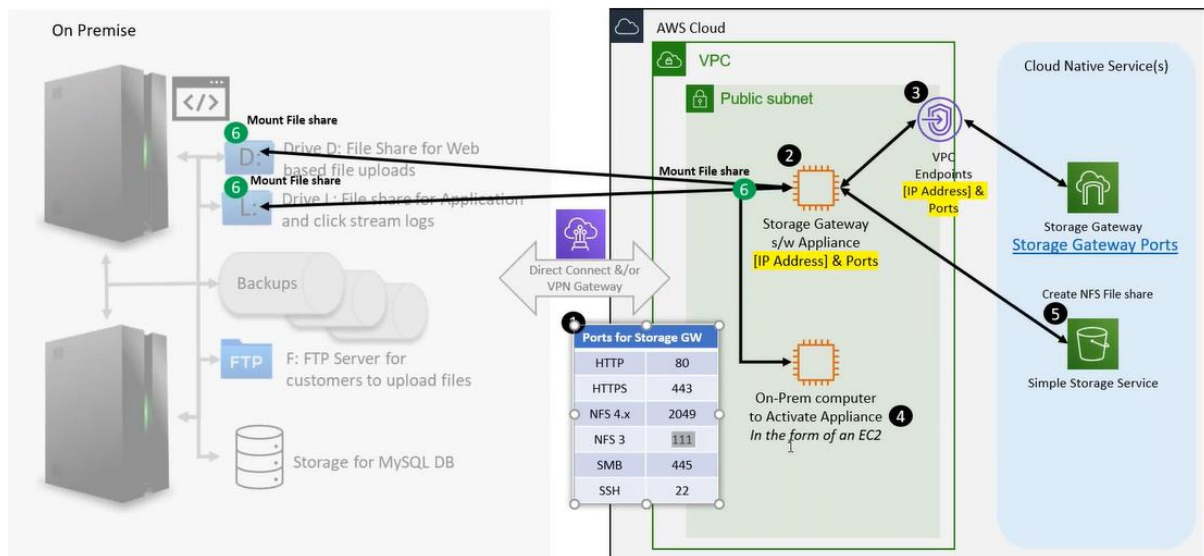
Description:

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop
HTTP	TCP	80	Custom 0.0.0.0/0, :::0	e.g. SSH for Admin Desktop
NFS	TCP	2049	Anywhere 0.0.0.0/0, :::0	e.g. SSH for Admin Desktop
SMB	TCP	445	Anywhere 0.0.0.0/0, :::0	e.g. SSH for Admin Desktop
HTTPS	TCP	443	Anywhere 0.0.0.0/0, :::0	e.g. SSH for Admin Desktop

step4: select service endpoint (VPC).

step4.1: create a vpc end point. (First enable DNS resolution/hostname for VPC).

This appliance will help us to convert the file storage to object storage.



Note: To connect the storage gateway appliance (VPC) <-> storage gateway (Cloud native) service, either connect through public network (option not secure), since VPC and Storage-Gateway service both are provided by AWS can be connected through VPC end point (private network, secure and fast).

step5: connect to gateway

ip of end point, ip of s/w appliance (private ip).

step6: Activate Gateway (Also add default sg)

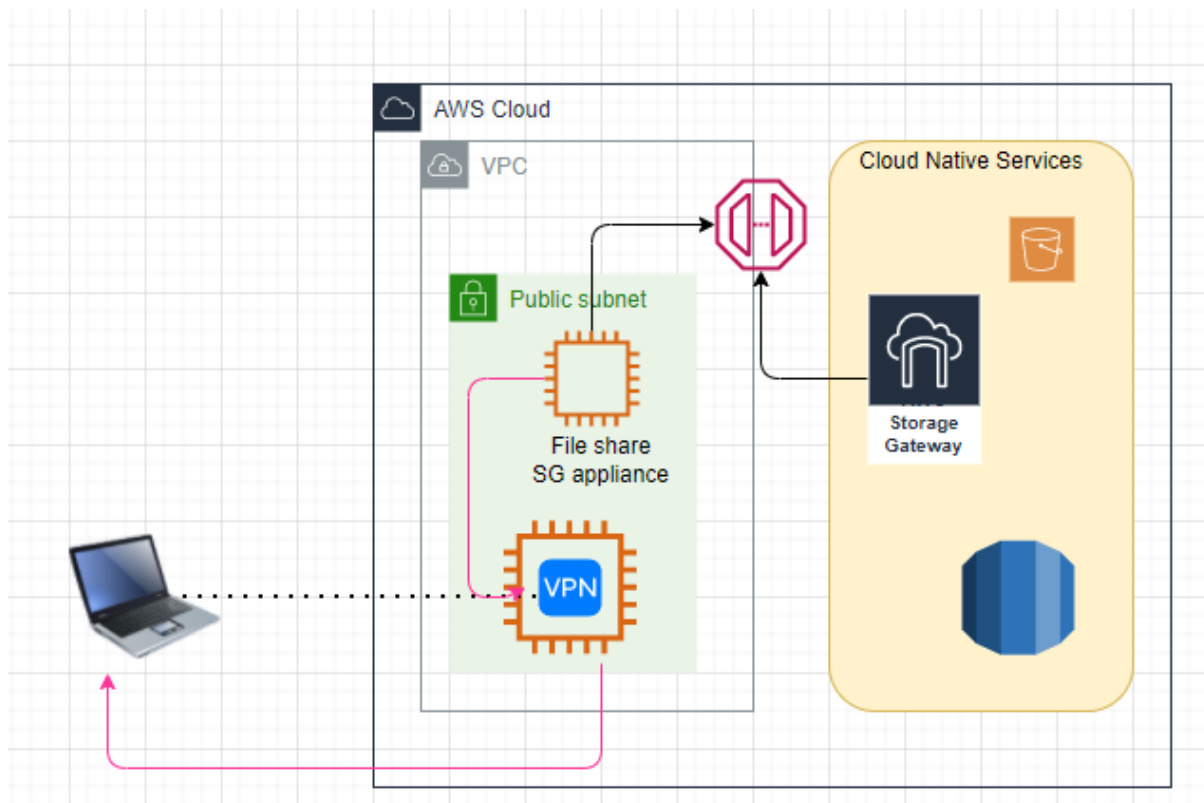
step7: configure local disks (cache memory on S/W appliance).

step8: logging

step9: Storage Gateway is ready; Storage Gateway is an appliance on top of that we create a file share.

step1: Create file share (S3)

step2: S3 file share available



step1: login to bastion-host (add sg default)

step2: mount on window/linux as D: and L:

step2.1: 7.9EB space available on windows

step3: copy data in D: will move to S3 bucket.

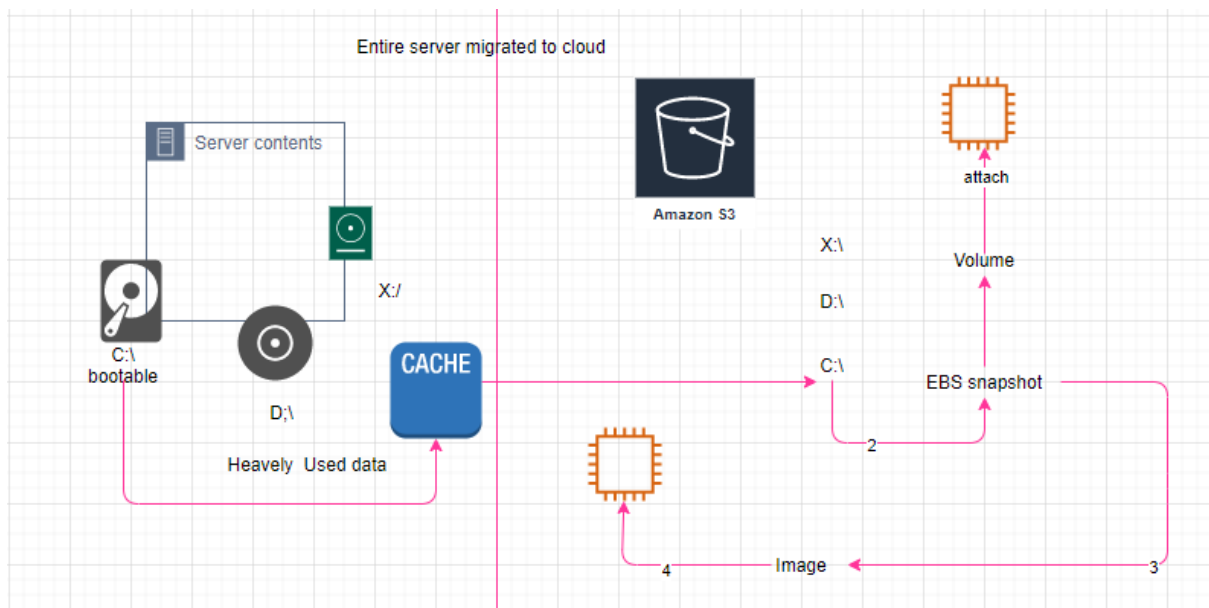
step4: copy file in S3 available in D:

Storage Gateway supports Volume Gateway (c:), Tape Gateway.

Tapes are for archive/backup purpose.

What is benefit of Volume Gateway?

C drive backed with volume share, can convert your server to EC2 image.



Data Sync:

S3 Transfer acceleration (access through the Edge location)

Beneficial if and only if your bucket is far away from your current location

As we enable the S3 Transfer Acceleration enabled, all the DC and POP must update their DNS entry.